

ApET: Approximation-Error Guided Token Compression for Efficient VLMs

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Abstract

Recent Vision-Language Models (VLMs) have demonstrated remarkable multimodal understanding capabilities, yet the redundant visual tokens incur prohibitive computational overhead and degrade inference efficiency. Prior studies typically relies on [CLS] attention or text-vision cross-attention to identify and discard redundant visual tokens. Despite promising results, such solutions are prone to introduce positional bias and, more critically, are incompatible with efficient attention kernels such as FlashAttention, limiting their practical deployment for VLM acceleration. In this paper, we step away from attention dependencies and revisit visual token compression from an information-theoretic perspective, aiming to maximally preserve visual information without any attention involvement. We present ApET, an **A**pproximation-**E**rror guided **T**oken compression framework. ApET first reconstructs the original visual tokens with a small set of basis tokens via linear approximation, then leverages the approximation error to identify and drop the least informative tokens. Extensive experiments across multiple VLMs and benchmarks demonstrate that ApET retains 95.2% of the original performance on image-understanding tasks and even attains 100.4% on video-understanding tasks, while compressing the token budgets by 88.9% and 87.5%, respectively. Thanks to its attention-free design, ApET seamlessly integrates with FlashAttention, enabling further inference acceleration and making VLM deployment more practical. Code is available at <https://github.com/MaQianKun0/ApET>.

1. Introduction

Recent advances in Vision-Language Models (VLMs) [4, 9, 11, 19, 22–24, 27, 28, 30, 31] have demonstrated their

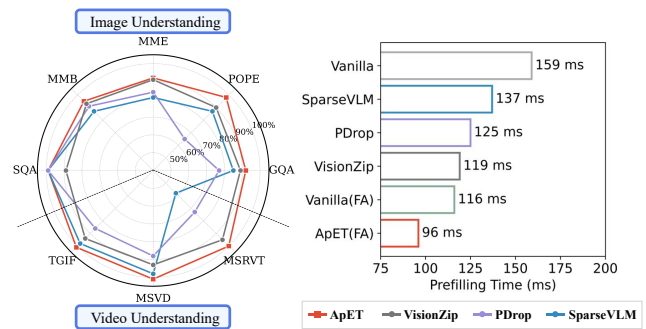


Figure 1. **Performance-Efficiency Comparison.** (Left) Performance across various image- and video-understanding benchmarks, ApET significantly outperforms existing token reduction approaches. (Right) ApET can be seamlessly combined with FlashAttention (FA) to further reduce prefilling time on Qwen-2.5-VL, whereas prior token reduction approaches are incompatible with FlashAttention.

exceptional capabilities in understanding and reasoning about visual content across a wide range of vision-language tasks. However, VLMs face significant challenges in computational efficiency and scalability, primarily due to the large number of visual tokens required to represent high-resolution images and long video sequences. These burdensome demands markedly constrain the practical deployment of VLMs in real-world scenarios [20, 33, 47, 49, 50].

Recognizing that many visual tokens are uninformative for VLM response, recent studies [10, 44, 46, 48, 53, 57] have introduced training-free token compression techniques that prune or merge less-informative tokens to improve efficiency without sacrificing performance. These methods typically use attention weights to identify and retain salient tokens. However, their inherent dependence on the attention mechanism still limits their practical implementation. Specifically, since these methods require access to attention weights to determine token importance, they are incompatible with efficient attention implementations such

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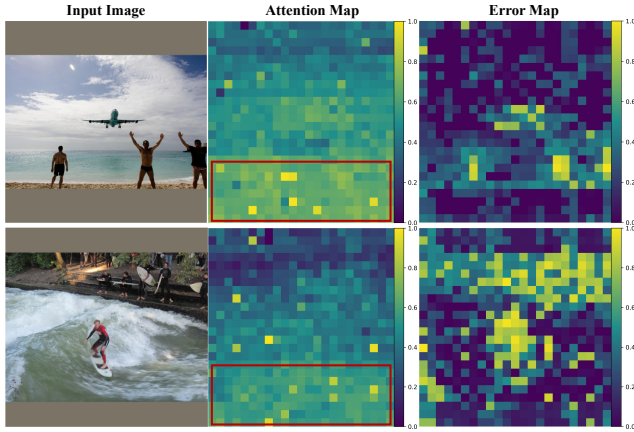


Figure 2. **Comparison of the attention map and approximate error map.** Attention-guided methods exhibit a positional bias that is agnostic to actual information content, assigning disproportionately high importance to later positions regardless of their semantic relevance (red boxes). In contrast, the error map provides an intuitive and content-aware reflection of token importance.

as FlashAttention (FA), which do not provide attention weights. Consequently, such compression techniques cannot be seamlessly integrated with these optimized attention operators.

To empirically demonstrate this, we conducted an efficiency comparison experiment (Figure 1 right). The results reveal that existing attention-guided compression methods are less efficient than directly employing FA. Moreover, since the use of FA incurs virtually no degradation in model performance, attention-guided compression approaches offer no practical advantage in real-world applications. Additionally, the visualization in Figure 2 reveals a positional bias inherent in the attention mechanisms of LLMs: visual tokens positioned closer to textual tokens (i.e., those appearing later in the sequence) consistently receive disproportionately high attention. This bias increases the risk that potentially important tokens will be erroneously discarded during compression, thereby degrading model performance.

To address the aforementioned challenges, we propose the **Approximation-Error guided Token reduction method** (ApET). ApET first linearly reconstructs each token using a carefully selected subset of tokens and then estimates its information content by computing the reconstruction error. This enables us to evaluate token importance without relying on external signals. Our key insight is that a token’s intrinsic information can be effectively captured by its linear approximation error: a small reconstruction error indicates that the token can be well represented by the selected subset, suggesting low information content, whereas a large error implies higher information content (see Section 3.2 for details). Unlike attention-guided methods, ApET focuses on the intrinsic characteristics of tokens, thereby inherently avoiding position bias, as illustrated in Figure 2. Further-

more, ApET not only achieves state-of-the-art performance on both image- and video-understanding tasks but also remains compatible with efficient attention kernels such as FA, enabling further computational efficiency, as demonstrated in Figure 1.

We summarize our contributions as follows:

- We conduct the first comprehensive analysis of visual token evaluation in VLMs from an information-theoretic perspective. By estimating the information carried by each token via its approximation error, we can assess token importance without relying on external signals, yielding a novel perspective on visual token compression in VLMs.
- Based on our exploration, we introduce ApET, an approximation-error guided token compression method that reconstructs tokens via linear approximation and quantifies their importance by the approximation-error, thereby eliminating positional bias while retaining compatibility with efficient operators.
- Extensive experiments show that ApET achieves outstanding performance on both image- and video-understanding benchmarks, while its fusion with FlashAttention yields an additional gain in model efficiency.

2. Related Work

Efficient Vision-Language Models. Leveraging powerful auto-regressive LLMs [1, 14, 26, 39], contemporary VLMs predominantly adopt an encoder-projector-decoder paradigm that fuses visual tokens with textual sequences [8, 12, 37, 38]. However, as image resolution or the number of frames increases, the volume of visual tokens increases proportionally, triggering a quadratic increase in computational cost due to the self-attention mechanism [7, 13, 40, 52] and hampering deployment in resource-constrained scenarios [6, 10, 21, 32, 54, 55]. To counteract this limitation, recent models embed specialized bottlenecks—e.g., Instruct-BLIP’s Q-Former [15] and OpenFlamingo’s perceiver resampler [2, 18]—that compress dense visual representations into compact feature sets before reaching the LLM decoder. Complementing these architectural refinements, FlashAttention [16, 17] reorders memory access patterns to exploit GPU hierarchies, yielding substantial speed-ups without sacrificing downstream accuracy.

Visual Token Compression for VLMs. To mitigate the quadratic complexity of self-attention introduced by dense vision tokens, existing approaches can be broadly categorized into two types: 1) *Text-guided compression methods* [29, 36, 45, 51, 57], which leverage textual queries to eliminate visual tokens that are irrelevant to the text. For example, SparseVLM [57] introduces an iterative sparsification strategy that selects visually relevant text tokens to assess the importance of vision tokens. Similarly, PyramidDrop [45] performs progressive pruning across multiple

decoding layers to balance computational efficiency with contextual integrity. 2) *Text-agnostic compression methods* [3, 34, 41, 42, 48, 56], which rely solely on self-attention among visual tokens and the [CLS] token to identify and prune redundant visual information. For instance, VisionZip [48] selects dominant tokens based on attention scores associated with the [CLS] token, while VisionDrop [46] filters significant tokens using self-attention weights between visual tokens. Despite their effectiveness, these methods typically depend on attention weights, which can conflict with the implementation of efficient attention mechanisms such as FlashAttention. In this work, we propose an information-theoretic approach that removes the least informative tokens without relying on attention weights. This ensures full compatibility with FlashAttention while achieving compression performance on par with existing methods.

3. Method

3.1. Preliminary

Most existing VLMs are composed of three core components: a visual encoder, a feature projector, and an LLM decoder. Taking an image as input, the visual encoder first processes the image patches, and the projector converts them into a set of n visual tokens, denoted as $V = \{v_1, \dots, v_n\}$. These visual tokens are then prepended to the tokenized textual query T and fed into the LLM decoder for autoregressive next-token generation. This process can be formally expressed as: $y_t \sim p_\theta(y_t|V, T, \mathbf{y}_{<t})$, where θ denotes the parameters of the VLM, y_t is the next token sampled from the output probability distribution $p_\theta(\cdot)$, and $\mathbf{y}_{<t}$ represents the sequence of tokens generated before timestep t .

3.2. Information-Theoretic Analysis

Unlike most previous approaches that rely on attention weights to assess token importance, we adopt an information-theoretic perspective, aiming to maximize the information content retained in compressed visual tokens. Specifically, given a set of visual tokens V , we sample a compressed subset S with the objective of maximizing the mutual information $I(V; S)$, as follows:

$$S^* = \arg \max_S I(V; S) \quad (1)$$

The objective is to identify an optimal subset $S \subseteq V$ that retains the maximum information about the original set V . According to information theory [35], the mutual information $I(V; S)$ can be defined as:

$$I(V; S) = H(V) - H(V|S) \quad (2)$$

Here, $H(V)$ denotes the entropy of V , while $H(V|S)$ quantifies the remaining uncertainty after observing S . Since

$H(V)$ is constant for a given input, maximizing $I(V; S)$ reduces to minimizing $H(V|S)$. In practice, however, this minimization is intractable: the joint distribution $p(V, S)$ required to compute $H(V|S)$ is unavailable.

Theorem. (Lower bound on the minimal reconstruction MSE ξ). Let $H(x|z)$ denote the conditional entropy of the input x given the intermediate feature z . The minimal reconstruction MSE ξ is bounded by: $\xi \geq \frac{1}{(2\pi e)} \exp(\frac{2H(x|z)}{d})$.

Leveraging this theorem, we can bound the conditional entropy in terms of the expected reconstruction mean-square error (MSE) [43]. Specializing the general bound to our setting yields:

$$\frac{1}{(2\pi e)} \exp(\frac{2H(V|S)}{d}) \leq \xi \quad (3)$$

According to Eq. 3, minimising the reconstruction MSE ξ is connected to minimizing the conditional entropy $H(V|S)$. Thus, the significance of visual tokens can be evaluated by quantifying the reconstruction error.

3.3. Approximation-Error Token Compression

Based on our analysis, reconstruction MSE serves as an effective indicator of the importance of visual tokens—larger errors correspond to greater token significance. To ensure simplicity and computational efficiency, we eschew the use of an additional reconstruction model. Instead, we adopt a linear approximation strategy, leveraging the resulting approximation error to evaluate token importance. Specifically, we first select a compact subset of tokens as base tokens and linearly reconstruct the remaining tokens from them. We then compute the per-token feature discrepancy between the original and reconstructed representations. Grounded in the preceding theoretical analysis, we adopt the magnitude of this reconstruction error as a principled surrogate for token importance. Building upon this, we propose the **Approximation-Error guided Token compression method**, as illustrated in Figure 3.

ApET can be seamlessly integrated at any layer of either the vision encoder or the LLM. In this work, we instantiate it immediately after the vision encoder and at an intermediate LLM layer (e.g., layer 16 in LLaVA). Concretely, ApET proceeds in three stages: token selection, ranking by approximation-error, and token merging.

Token Selection. Given a visual token set V^l at layer l , we construct M basis token subset $B = \{b_1, \dots, b_M\}$ via sampling, where the hyper-parameter M controls the sampling size. Three sampling strategies are investigated in this paper: 1) Farthest Distance Sampling (FPS), 2) Density Peak Clustering (DPC), and 3) Random Sampling. Ablation results are reported in Table 5.

Approximation-Error Computation. For each token $v \in V$, we seek a linear combination of the tokens in B that

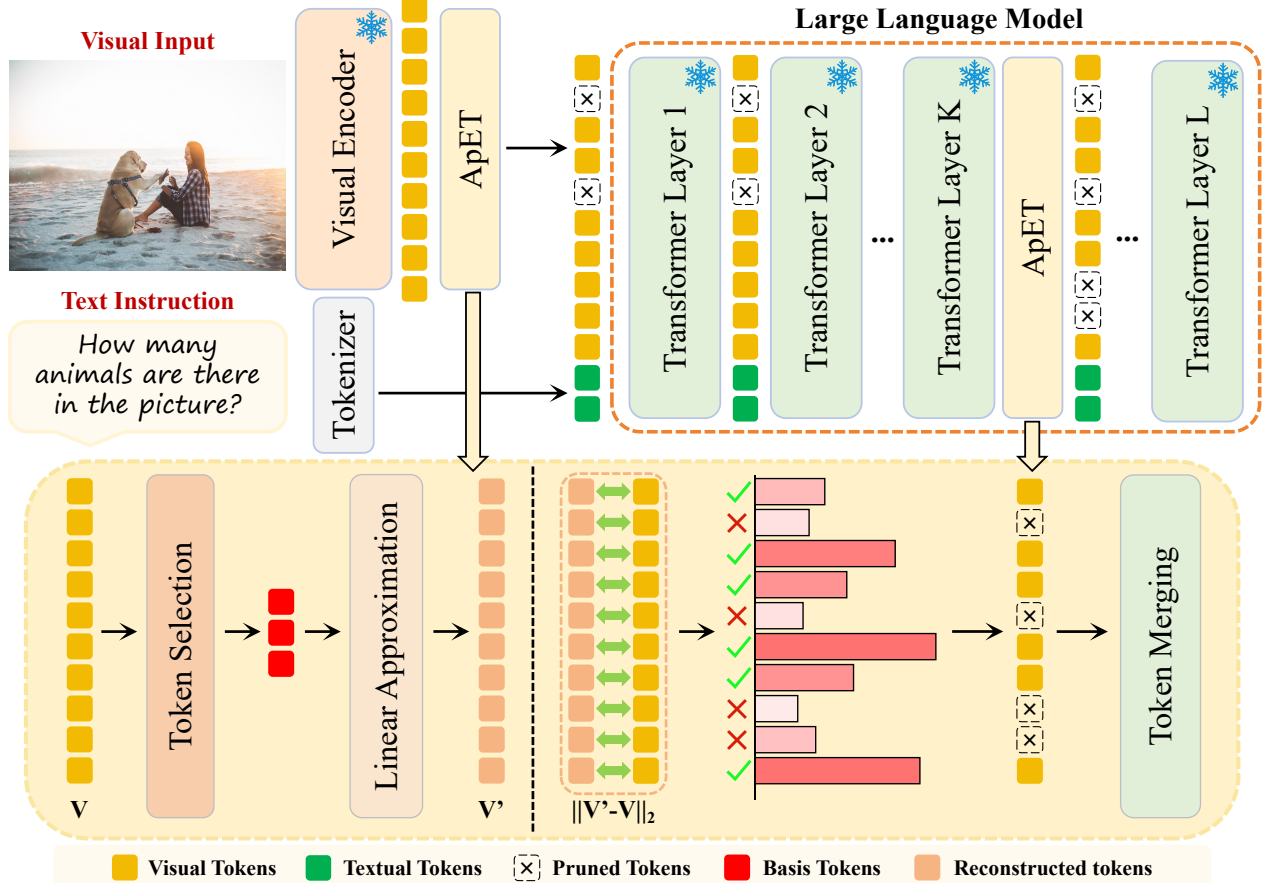


Figure 3. **The overview of ApET.** ApET selectively filters tokens exhibiting minimal information content based on approximation error, and subsequently employs token merging strategies to accomplish token compression.

approximates v :

$$v' \approx \sum_{i=1}^M \alpha_i b_i \quad (4)$$

Here, the coefficients α_i are assembled into a matrix $A \in \mathbb{R}^{M \times N}$, so that the entire set V is approximated by the single relation: $V \approx BA$. This coefficient matrix A is obtained by solving the corresponding linear system. After that, the approximate error used to measure the information of a token is defined as:

$$\xi = \Delta v = \|v - v'\|_2 \quad (5)$$

The approximation error can be employed to evaluate token significance while remaining compatible with FA.

Token Merging. Based on approximation error, we rank the importance of visual tokens and selectively reduce those with lower significance. It is worth noting that, in order to avoid omitting any potentially important tokens, we explicitly retain the base tokens as part of the significant token set. To mitigate potential information loss, we propose a similarity-based token merging strategy. Specifically, for each token marked for removal, we identify and merge it

with the most similar token among those retained. After all unselected tokens have been assigned to their closest counterparts, we apply average merging [6] within each group to generate merged tokens.

4. Experiments

4.1. Experimental Settings

Models and Baselines. To validate the effectiveness of our proposed method, we apply ApET to four representative VLMs. For image understanding tasks, we integrate our approach into the widely adopted LLaVA family, specifically LLaVA-1.5-7B [27] and LLaVA-Next-7B [28]. To further assess the generalizability of our method, we extend the evaluation to the more advanced Qwen2.5-VL-7B model [5]. In the context of video understanding, we adopt Video-LLaVA-7B [25] as the baseline model and evaluate our approach on three commonly used video benchmarks. We compare the performance of ApET against several state-of-the-art visual token reduction methods, including ToMe [6], FastV [10], SparseVLM [57], PDrop [45], and VisionZip [48].

Table 1. **Performance of ApET on LLaVA-1.5 across 9 image understanding benchmarks.** The original number of visual tokens is 576. The last column reports the average accuracy relative to the theoretical upper bound. The best result in each setting is highlighted.

Method	GQA	MMB	MMB ^{CN}	MME	POPE	SQA	VQA ^{V2}	VQA ^{Text}	VizWiz	Avg.
<i>Upper Bound, 576 Tokens (100%)</i>										
Vanilla	61.9	64.7	58.1	1862	85.9	69.5	78.5	58.2	50.0	100%
<i>Retain 192 Tokens (↓ 66.7%)</i>										
ToMe (ICLR23)	54.3	60.5	-	1563	72.4	65.2	68.0	52.1	-	88.5%
FastV (ECCV24)	52.7	61.2	57.0	1612	64.8	67.3	67.1	52.5	50.8	90.4%
SparseVLM (ICML25)	57.6	62.5	53.7	1721	83.6	69.1	75.6	56.1	50.5	96.1%
PDrop (CVPR25)	57.3	63.3	56.8	1797	82.3	69.0	75.1	56.5	51.1	97.2%
VisionZip (CVPR25)	59.3	63.0	-	1783	85.3	68.9	77.4	57.3	-	97.8%
ApET (Ours)	60.2	63.4	57.9	1808	86.3	68.5	76.2	54.4	50.2	98.0%
<i>Retain 128 Tokens (↓ 77.8%)</i>										
ToMe (ICLR23)	52.4	53.3	-	1343	62.8	59.6	63.0	49.1	-	80.4%
FastV (ECCV24)	49.6	56.1	56.4	1490	59.6	60.2	61.8	50.6	51.3	85.4%
SparseVLM (ICML25)	56.0	60.0	51.1	1696	80.5	67.1	73.8	54.9	51.4	93.7%
PDrop (CVPR25)	57.1	61.6	56.6	1761	82.3	68.4	72.9	56.6	51.0	96.2%
VisionZip (CVPR25)	57.6	62.0	-	1763	83.2	68.9	75.6	56.8	-	96.2%
ApET (Ours)	58.9	62.3	56.4	1801	86.1	68.7	75.1	53.9	50.7	97.1%
<i>Retain 64 Tokens (↓ 88.9%)</i>										
ToMe (ICLR23)	48.6	43.7	-	1138	52.5	50.0	57.1	45.3	-	70.1%
FastV (ECCV24)	46.1	48.0	52.7	1256	48.0	51.1	55.0	47.8	50.8	76.7%
SparseVLM (ICML25)	52.7	56.2	46.1	1505	75.1	62.2	68.2	51.8	50.1	87.2%
PDrop (CVPR25)	47.5	58.8	50.5	1561	55.9	69.2	69.2	50.6	50.7	86.6%
VisionZip (CVPR25)	55.1	60.1	-	1690	77.0	69.0	72.4	55.5	-	92.7%
ApET (Ours)	56.9	61.2	54.4	1714	84.4	68.9	72.5	53.0	51.9	95.2%

Implementation Details. We adopt the standard inference configurations provided in the official implementations of each VLM. Our method can be seamlessly integrated into these models in a training-free manner. Specifically, for each VLM, we perform visual token compression at the output of the visual encoder and within a middle layer of the LLM: layer 16 for the LLaVA-series models and layer 14 for Qwen2.5-VL.

4.2. Main Results

Image Understanding Tasks. As shown in Table 1, we integrate ApET to LLaVA-1.5-7B and compare its performance with existing baselines across 9 image understanding tasks. Following the setup in [48, 57], we compress the visual token sequence to 192, 128 and 64 tokens to evaluate the advantages of our method. Our method consistently outperforms all competing methods across different token reduction rates, with its superiority becoming increasingly pronounced at higher compression ratios. Specifically, even

under an aggressive reduction rate of 88.9%, it’s average performance outperforms the second-best VisionZip by a substantial margin of 2.5%.

To further validate the effectiveness of our approach, we extend our evaluation to LLaVA-NeXT-7B. Unlike LLaVA-1.5, LLaVA-NeXT decomposes each image into four parts together with a down-scaled full-view, yielding five distinct images and consequently a markedly larger visual token budget, making visual token compression more imperative. Following the experimental protocol in [48], we conducted comparative experiments under three vision token budgets (640, 320, and 160). As shown in Table 2, our method consistently exhibits superior performance across all three configurations. Specifically, using only 160 tokens, our method achieves 97.6% average accuracy, attaining the best performance on 4 out of 7 benchmarks.

Furthermore, we evaluate the competing algorithms on the more capable model, that is, Qwen-2.5-VL-7B. Unlike previous work, it adopts dynamic resolution, so each im-

Table 2. Performance comparison of ApET on LLaVA-NeXT-7B across multiple image understanding benchmarks.

Method	GQA	MMB	MMB ^{CN}	MME	POPE	VQA ^{V2}	VQA ^{Text}	Avg.
<i>Upper Bound, 2880 Tokens (100%)</i>								
Vanilla	64.2	67.4	60.6	1851	86.5	81.8	61.3	100%
<i>Retain 640 Tokens (↓ 77.8%)</i>								
SparseVLM	60.3	65.8	58.5	1773	84.2	77.1	57.8	95.7%
PDrop	60.6	65.5	58.5	1781	83.7	78.3	57.4	95.8%
VisionZip	61.3	66.2	-	1787	85.9	79.1	60.2	97.4%
ApET (Ours)	63.0	65.3	59.3	1815	87.2	79.2	57.9	97.6%
<i>Retain 320 Tokens (↓ 88.9%)</i>								
SparseVLM	59.3	64.2	55.9	1690	83.3	75.7	58.8	93.7%
PDrop	56.4	63.4	56.2	1663	77.6	73.5	54.4	90.4%
VisionZip	59.3	63.1	-	1702	82.1	76.2	58.9	93.7%
ApET (Ours)	61.0	63.5	56.6	1783	85.6	75.8	54.4	94.2%
<i>Retain 160 Tokens (↓ 94.4%)</i>								
SparseVLM	51.2	52.1	48.6	1542	72.7	66.3	46.4	80.2%
PDrop	54.9	61.8	54.9	1513	72.3	70.2	52.7	86.4%
VisionZip	55.5	60.1	-	1628	74.8	71.4	56.2	88.2%
ApET (Ours)	58.4	60.8	52.3	1680	82.6	72.7	53.8	90.1%

age is encoded into a variable length sequence of visual tokens. The length is bounded by the min pixels and max pixels hyper-parameters, which we fix to 256 and 2 048, respectively, for all methods to ensure a fair comparison. Table 3 reports the results on five image-understanding benchmarks. ApET consistently surpasses all baseline at every retention rate, confirming its efficacy at high resolution and its seamless compatibility with variable-resolution inputs.

Notably, most baselines estimate token importance based on attention weights. Although this heuristic is simple and often effective, its weight-access pattern is fundamentally incompatible with FlashAttention. Our method, by contrast, not only matches or exceeds their accuracy, but also plugs seamlessly into FlashAttention, delivering both efficiency and scalability.

Video Understanding Tasks. To evaluate the proposed method’s effectiveness for video understanding, we integrated it into Video-LLaVA, where each video is encoded as 8 frames of 256 visual tokens (2048 tokens in total). Under the aggressive compression budget of retaining only 256 tokens (32 per frame), ApET achieves 100.7% averaged accuracy across the three standard benchmarks, surpassing the previous state-of-the-art VisionZip by a substantial 6.3%, as shown in Table 4. This exceptional result not only establishes a new state of the art, but also surpasses the performance of the original Video-LLaVA-7B model while preserving merely 12.5% of the original visual tokens. Such a finding provides strong empirical support for the hypothesis that video sequences contain a considerable proportion of redundant, distracting, or even misleading tokens. Consequently, token compression methods serve a dual purpose:

Table 3. Performance comparisons on Qwen-2.5-VL-7B across five image understanding benchmarks. †Evaluation is based on our re-implementation.

Method	GQA	POPE	SQA	MME	MMB	Avg.
<i>Upper Bound, 256~2048 Tokens (100%)</i>						
Vanilla	60.5	86.2	76.7	2327	83.3	100%
<i>Retain 20% Tokens in Average (↓ 80%)</i>						
SparseVLM†	54.7	73.6	71.6	2063	76.0	89.8%
PDrop†	55.1	78.4	70.9	2117	77.3	91.6%
VisionZip†	56.8	82.4	76.3	2134	79.3	95.2%
ApET (Ours)	58.8	85.1	74.9	2181	78.6	96.3%
<i>Retain 10% Tokens in Average (↓ 90%)</i>						
SparseVLM†	51.3	71.9	68.2	1849	71.7	84.5%
PDrop†	52.0	74.8	69.7	1886	73.6	86.6%
VisionZip†	52.4	78.9	74.1	2003	75.6	90.3%
ApET (Ours)	56.1	82.4	74.1	2055	72.6	92.1%

Table 4. Performance of ApET on Video-LLaVA-7B across three video understanding tasks. The original Video-LLaVA’s video token number is 2048, while we only retain 256 tokens for all methods.

Method	TGIF		MSVD		MSRVT		Avg.	
	Acc	Score	Acc	Score	Acc	Score	Acc	Score
Vanilla	46.9	3.34	69.8	3.91	57.1	3.49	100%	100%
FastV	44.2	3.29	60.3	3.72	40.6	3.18	83.9%	94.9%
	94.2%	98.5%	86.4%	95.1%	77.1%	91.1%		
SparseVLM	45.9	3.32	68.6	3.90	32.9	3.02	84.6%	95.2%
	98.9%	99.4%	98.3%	99.7%	57.6%	86.5%		
PDrop	40.3	3.21	61.5	3.74	41.8	3.19	82.4%	94.4%
	85.9%	96.1%	88.1%	95.7%	73.2%	91.4%		
VisionZip	44.3	3.29	65.2	3.83	54.5	3.43	94.4%	98.2%
	94.5%	98.5%	93.4%	98.0%	95.4%	98.3%		
ApET (ours)	47.6	3.37	70.4	3.94	56.9	3.47	100.7%	100.4%
	101.5%	100.9%	100.9%	100.8%	99.6%	99.4%		

they markedly enhance computational efficiency and simultaneously act as a denoising mechanism that removes spurious visual cues, thereby improving the model’s overall performance.

Furthermore, ApET demonstrates consistent superiority across all three video understanding benchmarks, whereas other methods display pronounced dataset-specific sensitivity. For example, VisionZip delivers competitive accuracy on TGIF and MSRVT yet incurs a performance drop on MSVD. In contrast, ApET maintains robust performance across every evaluation set, underscoring its stability and reliability in diverse video understanding scenarios. Closer analysis reveals that the performance gain afforded by ApET is even more pronounced for video than for image understanding. We attribute this phenomenon to the exacerbated positional bias inherent in attention-based pruning

Table 5. **Ablation study on sampling strategies across three image understanding tasks.** Experiments are performed with LLaVA-1.5-7B and Qwen2.5-VL-7B at token compression ratios of 11.1% and 10%, respectively. “Random” denotes random sampling, “DPC” indicates density peak clustering, and “FPS” refers to farthest point sampling.

Model	Sampling	GQA	POPE	MME
LLaVA-1.5-7B	Random	55.7	82.5	1634
	DPC	57.1	84.8	1691
	FPS	56.9	84.4	1714
Qwen2.5-VL-7B	Random	55.6	80.8	1944
	DPC	56.5	82.5	1998
	FPS	56.1	82.4	2055

strategies when they are confronted with long, temporally extended contexts. By eschewing any dependence on attention weights, ApET eliminates this bias entirely, enabling superior compression quality and, consequently, stronger generalization at the video level.

4.3. Ablation Studies

Effectiveness of Different Sampling Methods. The Token Selection operation in our method identifies base tokens for the linear approximation process. In Table 5, we evaluate the impact of three sampling strategies on ApET performance: 1) Random sampling, where M visual tokens are randomly selected as base tokens; 2) Density Peak Clustering (DPC), which derives M cluster centers from all visual tokens; 3) Farthest Point Sampling (FPS), which selects M maximally diverse tokens as base tokens. Experimental results indicate that both FPS and DPC achieve competitive performance. However, DPC requires computing M cluster centers from N visual tokens, incurring significant computational overhead when N is large—an inefficiency that hampers practical deployment. Thus, we adopt FPS as the default sampling strategy for ApET, striking an optimal balance between accuracy and efficiency. Notably, even under random sampling, ApET maintains robust performance, validating the efficacy of our core insight: leveraging approximation error as a proxy for token importance in attention mechanisms.

Analysis on Hyper-parameter. In our token selection operation, we select M visual tokens as base tokens for subsequent linear approximation. To assess the impact of varying M on model performance, we conducted an ablation study in Table 6. Experimental results indicate that performance is stable across a wide range of M , peaking at $M = 10$. We therefore adopt $M = 10$ for all experiments. Furthermore, we summarize two conclusions. 1) M is insensitive to the performance, which indicates that our method is robust to hyper-parameter settings. 2) M is vastly smaller than the total number of visual tokens N ($N = 576$ for LLaVA-1.5-7B), so the additional computational overhead introduced

Table 6. **Ablation studies of Hyper-parameter.** Experiments are performed with LLaVA-1.5-7B and Qwen2.5-VL-7B at token compression ratios of 11.1% and 10%, respectively.

Model	Settings	GQA	POPE	MME
LLaVA-1.5-7B	$M = 6$	56.0	83.5	1676
	$M = 8$	56.7	84.1	1705
	$M = 10$	56.9	84.4	1714
	$M = 12$	56.8	83.9	1699
	$M = 14$	56.4	83.7	1663
Qwen2.5-VL-7B	$M = 6$	55.9	81.7	2014
	$M = 8$	56.2	82.0	2047
	$M = 10$	56.1	82.4	2055
	$M = 12$	56.4	82.3	2044
	$M = 14$	55.7	81.5	1972

by the sampling or the approximation step is negligible.

Efficiency Analysis. Our proposed ApET significantly enhances inference efficiency in VLMs while maintaining performance on par with existing approaches. As shown in Table 7, we evaluate total inference time, prefilling time, and FLOPs on both LLaVA-1.5-7B and Qwen2.5-VL-7B, comparing ApET against several state-of-the-art methods. ApET consistently achieves optimal efficiency across both model architectures. Specifically, it yields a $1.46\times$ speedup in total inference time and a $1.38\times$ speedup in the prefilling time on LLaVA-1.5-7B. Notably, while competing methods yield efficiency improvements on LLaVA-1.5-7B, they underperform the baseline model on the more advanced Qwen2.5-VL-7B. This degradation stems from their reliance on attention weights to assess token importance. While LLaVA supports the extraction of these weights, Qwen2.5-VL does not, necessitating a costly recomputation of attention scores. This overhead introduces significant computational burden, ultimately reducing overall efficiency. In contrast, ApET estimates token importance via linear approximations applied directly to token representations, obviating any dependence on model internal intermediate results. This design enables seamless adaptation to architectures such as Qwen2.5-VL without additional compute. These results validate our central contribution: a model-agnostic token compression mechanism. We argue that effective compression strategies should be decoupled from model-specific internals, eschewing reliance on any architecture dependent computations—to ensure robust generalization across diverse VLMs.

Visualization. To further illustrate the effectiveness of our proposed ApET method, we provide qualitative visualizations of token selection results in Figure 4. These examples are drawn from the GQA dataset and showcase representative cases where attention-based token pruning methods fail to preserve task-relevant visual information, leading to incorrect predictions. In contrast, ApET guided by approximation error, successfully retains informative tokens

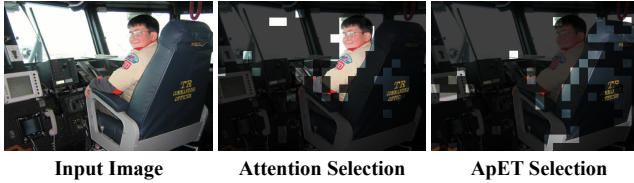
Table 7. **Efficiency analysis of ApET on LLaVA-1.5-7B and Qwen2.5-VL-7B.** We report total inference time, prefilling time, and TFLOPs; Δ indicates the speedup ratio. All measurements are performed on a single A100 GPU using the POPE benchmark.

Methods	Token	Total Time↓	$\Delta\uparrow$	Prefilling Time↓	$\Delta\uparrow$	TFLOPs
LLaVA-1.5-7B	100%	14:47	1.0×	86.9ms	1.0×	8.82
+ SparseVLM	11.1%	13:32	1.09×	77.8ms	1.12×	2.31
+ PDrop	11.1%	11:15	1.31×	69.2ms	1.26×	2.16
+ VisionZip	11.1%	10:36	1.39×	64.4ms	1.35×	2.03
+ ApET (ours)	11.1%	10:08	1.46×	62.8ms	1.38×	2.09

Methods	Token	Total Time↓	$\Delta\uparrow$	Prefilling Time↓	$\Delta\uparrow$	TFLOPs
Qwen2.5-VL-7B	100%	34:14	1.0×	158.5ms	1.0×	14.52
+ SparseVLM	10%	33:36	1.02×	136.8ms	1.16×	5.97
+ PDrop	10%	32:20	1.06×	125.5ms	1.26×	5.85
+ VisionZip	10%	30:06	1.14×	119.3ms	1.33×	5.68
+ ApET (ours)	10%	26:18	1.30×	104.7ms	1.51×	5.70

Query: Is this helicopter on or off?

Answer: Off



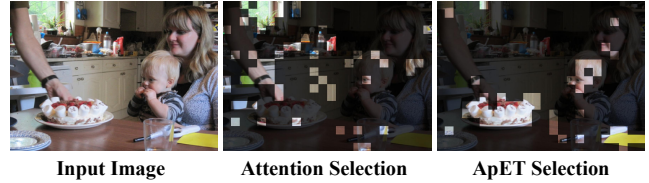
Query: What type of material is the crosswalk made of?

Answer: Concrete



Query: What type of food is to the left of the baby?

Answer: Dessert



Query: Which color is the cake?

Answer: White

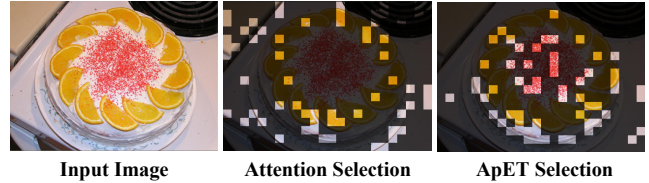


Figure 4. **Visualization of token compression.** We present four representative failure cases in which attention-driven token selection misguides the final prediction. For each case, we visualize the input question and its ground-truth answer, the original image, the subset of visual tokens preserved when ranking by attention weights, and the subset preserved by the approximation-error criterion proposed in this work. The retained tokens are highlighted for clear comparison.

and produces accurate answers. Each visualization includes the input image, the textual question, the ground-truth answer, and two sets of preserved tokens: one selected by attention weight based ranking (left), and the other by our approximation-error criterion (right). The retained tokens are highlighted on the image for clarity. For instance, in the first example, the query asks whether a helicopter is "on or off". Attention-based pruning fails to retain visual tokens corresponding to the cockpit indicators, resulting in an incorrect prediction. ApET, however, preserves tokens that capture the static state of the helicopter, enabling the model to infer the correct answer.

These visualizations underscore a key insight: approximation error serves as a more content-aware metric for token importance, unlike attention weights which are susceptible to positional and contextual biases. By focusing on reconstructability, ApET inherently prioritizes tokens that carry distinctive and irreplaceable visual information, leading to more robust and interpretable token reduction.

5. Conclusion

Vision-language models (VLMs) demand prohibitive computational overhead because of the massive number of visual tokens. While existing token compression methods that rely on attention weights not only introduce positional bias but also fail to be compatible with efficient attention kernels. In this work, we present ApET, an attention-free token compression framework that leverages approximation error to evaluate visual token importance from an information-theoretic perspective. By reconstructing tokens via linear approximation and retaining those with high reconstruction error, ApET preserves informative content while eliminating positional bias inherent in attention-based methods. Extensive experiments demonstrate that ApET achieves strong performance across both image and video understanding benchmarks, even under aggressive compression rates. Notably, it is fully compatible with efficient attention implementations like FlashAttention, enabling practical acceleration in real-world deployments.

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